Series resosnant circuit	*Parallel resosnant circuit*
1. An acceptor circuit	A rejector circuit
2. Resosnant frequency $v_r = \frac{1}{2\pi\sqrt{LC}}$	Resonant frequency $v_r = \frac{1}{2\pi\sqrt{LC}}$
3. At resonance the impedance is a minimum equal to the resistance in the circuit.	At resonance the impedance is maximum nearly equal to infinity
4. Selective	Selective
5. Used in the tuning circuit to separate the wanted frequency from the incoming frequencies by offering low impedance at that frequency.	Used to present a maximum impedance to the wanted frequency, usually in the plate circuit of valves.

Example 1. *A coil of self inductance 2 milli-henry and resistance 15 ohm is connected in parallel with a capacitance of 0.001* μ*F. Find* (*a*) *the frequency at which the current from an a.c. source to this circuit is minimum*, (*b*) *the peak-value of this make-up current if the peak value of supply voltage is 2 volt.*

Solution. (*a*) Current is minimum at resonant frequency.

$$v_0 = \frac{1}{2\pi}\left(\frac{1}{LC} - \frac{R^2}{L^2}\right)^{1/2} = \frac{1}{2\pi}\left[\frac{1}{(2\times10^{-3})\,(0.001\times10^{-6})} - \frac{(15)^2}{(2\times10^{-3})^2}\right]^{1/2} = 112590 \text{ Hz}$$

Aliter.

$$v_0 = \frac{1}{2\pi}\frac{1}{\sqrt{(LC)}}\left(\text{as } \frac{R^2}{L^2} << \frac{1}{LC}\right) = \frac{1}{2\times3.14\times\sqrt{(2\times10^{-3}\times0.001\times10^{-6})}} = 112596 \text{ Hz}$$

(*b*) Peak-value of make-up current at resonance

$$= \frac{\text{peak value of applied e.m.f.}}{\text{impedance at resonance}} = \frac{E_0}{L/CR} = \frac{E_0 CR}{L}$$

$$= \frac{2\times(0.001\times10^{-6})\times15}{2\times10^{-3}} = 15\times10^{-6}\,A$$

13.5 POWER IN AC CIRCUIT CONTAINING RESISTANCE, INDUCTANCE AND CAPACITANCE

Consider an ac circuit containing resistance, inductance and capacitance. E and I vary continuously with time. Therefore power is calculated at any instant and then its mean is calculated over a complete cycle.

The instantaneous values of the voltage and current are given by

$$E = E_0 \sin \omega t,$$

$$I = I_0 \sin (\omega t - \phi)$$

where ϕ is the phase difference between current and voltage.

Hence power at any instant is

$$E \times I = E_0 I_0 \sin \omega t \sin (\omega t - \phi)$$

$$= \frac{1}{2} E_0 I_0 [\cos \phi - \cos (2\omega t - \phi)] \quad ...(1)$$

Average power consumed over one complete cycle is

$$P = \frac{\int_0^T E\ I\ dt}{\int_0^T dt}$$

$$= \frac{\int_0^T \frac{1}{2} E_0\ I_0 [\cos \phi - \cos (2\omega t - \phi)]\, dt}{T}$$

$$= \frac{1}{2} \frac{E_0 I_0}{T} \left[(\cos \phi)\, t - \frac{\sin (2\omega t - \phi)}{2\omega} \right]_0^T$$

$$= \frac{1}{2} \frac{E_0 I_0}{T} \left[(\cos \phi)\, T - 0 - \frac{\sin (2\omega T - \phi)}{2\omega} + \frac{\sin (-\phi)}{2\omega} \right]$$

Now, $T = \frac{2\pi}{\omega}$ and $\sin (4\pi - \phi) = \sin (-\phi)$

$$P = \frac{1}{2} \frac{E_0 I_0 \omega}{2\pi} \left[(\cos \phi) \frac{2\pi}{\omega} - \frac{\sin (-\phi)}{2\omega} + \frac{\sin (-\phi)}{2\omega} \right]$$

$$= \frac{1}{2} E_0 I_0 \cos \phi$$

$$= \frac{E_0}{\sqrt{2}} \times \frac{I_0}{\sqrt{2}} \times \cos \phi$$

$$= E_{r.m.s.}\ I_{r.m.s.} \cdot \cos \phi \quad \ldots(2)$$

average power = (virtual volts) × (virtual amperes) × cos ϕ

The term (*Virtual volts* × *Virtual amperes*) *is called* apparent power *and cos* ϕ *is called the power factor. Thus*

True power = apparent power × power factor.

Evidently, the *power factor is the ratio of the true power to the apparent power.*

As cos ϕ is the factor by which the product of the r.m.s. values of the voltage and current must be multiplied to give the power dissipated, it is known as the 'power factor' of the circuit. For a circuit containing resistance, capacitance and inductance in series,

$$\tan \phi = \frac{\omega L - \frac{1}{\omega C}}{R}$$

From Fig. 13.24, the expression for the power factor is

$$\cos \phi = \frac{R}{\sqrt{\left\{ R^2 + \left(\omega L - \frac{1}{\omega C} \right)^2 \right\}}}$$

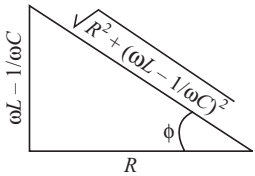


FIG. 13.24

Special cases.

(1) In a purely *resistive circuit*, ϕ = 0 or cos ϕ = 1,

∴ true power = $E_v \times I_v$

(2) In a *purely inductive circuit*, current lags behind the applied emf by 90° so that $\phi = 90°$ or $\cos \phi = 0$.

Thus true power consumed = 0

(3) In a *purely capacitive circuit*, current leads the applied voltage by 90° so that $\phi = -90°$ or $\cos (-90°) = \cos 90° = 0$

$\therefore$ true power = 0

(4) In an ac circuit containing a resistance and inductance in series.

$$\textit{Power factor}, \cos \phi = \frac{R}{\sqrt{R^2 + (L\omega)^2}}$$

(5) In an ac circuit containing a capacitance C and a resistance R in series,

$$\cos \phi = \frac{R}{\sqrt{\left(\frac{1}{C^2\omega^2} + R^2\right)}}$$

Wattless current

The average power dissipated during a complete cycle is $E_v I_v \cdot \cos \phi$

The current in A.C. circuit is said to be wattless when the average power consumed in the circuit is zero.

If an ac circuit is purely inductive or purely capacitive with no ohmic resistance, phase angle $\phi = \pi/2$ so that $\cos \phi = 0$ or the power consumed is zero. The current in such a circuit does not perform any useful work and is rightly called the *wattless* or *idle current*. In this situation, the circuit does not consume any power, though it offers a resistance to the flow of alternating current in it. It is the principle of choke coil.

Example 1. *An alternating voltage of 10 volts at 100 Hz is applied to a choke of inductance 5 henry and of resistance 200 ohms. Find the power factor of the coil and the power absorbed.*

Solution. Here, $E_v = 10\ V$, $v = 100\ Hz$, $L = 5\ H$, and $R = 200\ \Omega$

$$\text{Power factor}, \cos \phi = \frac{R}{\sqrt{R^2 + \omega^2 L^2}} = \frac{200}{\sqrt{(200)^2 + (2\pi \times 100 \times 5)^2}} = 0.062$$

$$\text{Power absorbed} = E_v \cdot I_v \cdot \cos \phi = E_v \cdot \frac{E_v}{Z} \cos \phi$$

$$= \frac{(E_v)^2}{\sqrt{R^2 + \omega^2 L^2}} \cos \phi = \frac{10 \times 10\ (0.062)}{\sqrt{(200)^2 + (2\pi \times 100 \times 5)^2}} = 0.00189 \text{ W}$$

Example 2. *A source of e.m.f. 50 V, r.m.s. is connected across an air cored coil. When the supply frequency is 50 Hz the power consumed is found to be 100 W, whereas when the frequency is increased to 100 Hz, the power becomes 50 W. Find the self-inductance and the resistance of the coil.*

Sol. Let the inductance and resistance required be L and R respectively.

$$I_v = \frac{E_v}{\sqrt{(R^2 + \omega^2 L^2)}}$$

When the supply frequency is 50 Hz, $\omega = 2\pi \times 50 = 100\ \pi$

$$I_v' = \frac{50}{\sqrt{R^2 + (100\pi)^2 L^2}}$$

The power consumed is dissipated only in the circuit resistance. It is,

$$I_v'^2\ R = \frac{50^2}{[R^2 + (100\pi)^2\ L^2]} R = 100 W \quad \ldots(1)$$

Let I_v'' be the current when the frequency is 100 Hz. The power consumed now is

$$I_v''^2\ = \frac{50^2 R}{[R^2 + (200\pi)^2 L^2]} = 50 W \quad \ldots(2)$$

From Eq. (1), $2500\ R = 100\ [R^2 + 10^4\pi^2 L^2]$ …(3)

From Eq. (2), $2500\ R = 50\ [R^2 + 4 \times 10^4\ \pi^2 L^2]$ …(4)

From Eq. (3) and (4), $R^2 = 2 \times 10^4 \times \pi^2 L^2$

or $R = 100\sqrt{2}\pi L$ ...(5)

Substituting for L^2 in Eq. (3),

$$2500\ R = 100\left[R^2 + \frac{10^4\pi^2 R^2}{2 \times 10^4 \times \pi^2}\right]$$

$$R = 16.7\ \Omega$$

From Eq. (5), $L = \frac{R}{100\sqrt{2}\pi} = \frac{16.7}{100\sqrt{2}\pi} = 0.0373\ H = 37.3\ mH$

Example 3. *What do you understand by wattless and power components of an alternating current*?

A coil of self-inductance 0.7 H is joined in series with a non-inductive resistance of 50 Ω. Calculate the wattless and power components as well as the total current when connected to a supply of 200 volt at a frequency of 50 Hz.

Solution. In a circuit containing an inductance L and a resistance R in series, the current lags behind the emf by an angle ϕ, where $\tan\phi = \omega L/R$.

Let the vector E represent the rms value of emf and the vector I the rms value of current (Fig. 13.25)

Component of I parallel to $E = I\cos\phi$

Component of I perpendicular to $E = I\sin\phi$

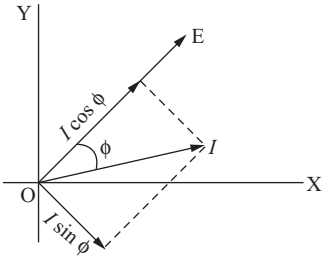


FIG. 13.25

The component $I\cos\phi$, which is in phase with the emf is responsible for the power dissipated in the circuit and is called the *power* component of the current. The component $I\sin\phi$, which lags 90° in phase behind the emf does not contribute to the power and is called the *idle* or *wattless* component of the current.

Here, $L = 0.7$ H, $R = 50\ \Omega$, $E_{rms} = 200\ V$, $\omega = 2\pi v = 2\pi \times 50$

$$\text{Impedance, } Z = \sqrt{(R^2 + \omega^2 L^2)} = \sqrt{[(50)^2 + (2\pi \times 50)^2\ (0.7)^2]} = 225.6\ \Omega$$

$$\therefore \quad \text{current, } I_{rms} = \frac{E_{rms}}{Z} = \frac{200}{225.6} = 0.89\ \text{A}$$

Power component of current is

$$I_{rms}\cos\phi = I_{rms} \times \frac{R}{\sqrt{(R^2 + \omega^2 L^2)}} = 0.89 \times \frac{50}{225.6} = 0.20\ \text{A}$$